



Virtualization Best Practices

1. Virtualization scenarios

Virtualization offers a lot of capabilities for your environment. It can be used in multiple scenarios. To get more details about it, refer to the *Virtualization Guide*, and in particular to the following sections:

- *the section called “Server consolidation”*
- *the section called “Virtualization benefits”*

This best practice guide provides advice for making the right choice in your environment. It recommends or discourages the use of options depending on your workload. Fixing configuration issues and performing tuning tasks increases the performance of VM Guests near to bare metal.

2. Before you apply modifications

2.1. Back up first

Changing the configuration of the VM Guest or the VM Host Server can lead to data loss or an unstable state. It is really important that you do backups of files, data, images, etc. before making any changes. Without backups, you cannot restore the original state after a data loss or a misconfiguration. Do not perform tests or experiments on production systems.

2.2. Test your workloads

The efficiency of a virtualization environment depends on many factors. This guide helps to make good choices when configuring virtualization in a production environment. Nothing is *carved in stone*. Hardware, workloads, resource capacity, etc. should all be considered when planning, testing and deploying your virtualization infrastructure. Testing your virtualized workloads is vital to a successful virtualization implementation.

3. Recommendations

3.1. Prefer the `libvirt` framework

SUSE strongly recommends using the `libvirt` framework to configure, manage and operate VM Host Servers and VM Guest. It offers a single interface (GUI and shell) for all supported virtualization technologies and therefore is easier to use than the hypervisor-specific tools.

We do not recommend using `libvirt` and hypervisor-specific tools at the same time, because changes done with the hypervisor-specific tools may not be recognized by the `libvirt` tool set. See *Chapter 8, `libvirt` daemons* for more information on `libvirt`.

3.2. `qemu-system-i386` compared to `qemu-system-x86_64`

Similar to real 64-bit PC hardware, `qemu-system-x86_64` supports VM Guests running a 32-bit or a 64-bit operating system. Because `qemu-system-x86_64` usually also provides better performance for 32-bit guests, SUSE recommends using `qemu-system-x86_64` for both 32-bit and 64-bit VM Guests on KVM. Scenarios where `qemu-system-i386` performs better are not supported by SUSE.

Xen also uses binaries from the `qemu` package but prefers `qemu-system-i386`, which can be used for both 32-bit and 64-bit Xen VM Guests. To maintain compatibility with the upstream Xen Community, SUSE encourages using `qemu-system-i386` for Xen VM Guests.

4. VM Host Server configuration and resource allocation

Allocation of resources for VM Guests is a crucial point when administrating virtual machines. When assigning resources to VM Guests, be aware that overcommitting resources may affect the performance of the VM Host Server and the VM Guests. If all VM Guests request all their resources simultaneously, the host needs to provide them all. If not, the host's performance is negatively affected and this in turn also has negative effects on the VM Guest's performance.

4.1. Memory

Linux manages memory in units called pages. On most systems the default page size is 4 KB. Linux and the CPU need to know which pages belong to which process. That information is stored in a page table. If a lot of processes are running, it takes more time to find where the memory is mapped, because of the time required to search the page table. To speed up the search, the TLB (Translation Lookaside Buffer) was invented. But on a system with a lot of memory, the TLB is not enough. To avoid any fallback to normal page table (resulting in a cache miss, which is

time consuming), huge pages can be used. Using huge pages will reduce TLB overhead and TLB misses (pagewalk). A host with 32 GB ($32 \times 1024 \times 1024 = 33,554,432$ KB) of memory and a 4 KB page size has a TLB with $33,554,432 / 4 = 8,388,608$ entries. Using a 2 MB (2048 KB) page size, the TLB only has $33,554,432 / 2048 = 16,384$ entries, considerably reducing the TLB misses.

4.1.1. Configuring the VM Host Server and the VM Guest to use huge pages

The AMD64/Intel 64 CPU architecture supports larger pages than 4 KB: huge pages. To determine the size of huge pages available on your system (could be 2 MB or 1 GB), check the `flags` line in the output of `/proc/cpuinfo` for occurrences of `pse` and/or `pdpe1gb`.

Table 1. Determine the available huge pages size

CPU flag	Huge pages size available
Empty string	No huge pages available
pse	2 MB
pdpe1gb	1 GB

Using huge pages improves the performance of VM Guests and reduces host memory consumption.

By default, the system uses THP. To make huge pages available on your system, activate it at boot time with `hugepages=1`, and—optionally—add the huge pages size with, for example, `hugepagesz=2MB`.



1 GB huge pages

1 GB pages can only be allocated at boot time and cannot be freed afterward.

To allocate and use the huge page table (HugeTlbPage), you need to mount `hugetlbfs` with correct permissions.



Restrictions of huge pages

Even if huge pages provide the best performance, they do come with some drawbacks. You lose features such as Memory ballooning (see *the section called “virtio balloon”*),

KSM (see the section called “KSM and page sharing”), and huge pages cannot be swapped.

Procedure 1. Configuring the use of huge pages

1. Mount `hugetlbfs` to `/dev/hugepages`:

```
>sudo mount -t hugetlbfs hugetlbfs /dev/hugepages
```

2. To reserve memory for huge pages, use the `sysctl` command. If your system has a huge page size of 2 MB (2048 KB), and you want to reserve 1 GB (1,048,576 KB) for your VM Guest, you need $1,048,576/2048=512$ pages in the pool:

```
>sudo sysctl vm.nr_hugepages=512
```

The value is written to `/proc/sys/vm/nr_hugepages` and represents the current number of *persistent* huge pages in the kernel's huge page pool. *Persistent* huge pages are returned to the huge page pool when freed by a task.

3. Add the `memoryBacking` element in the VM Guest configuration file (by running `virsh edit CONFIGURATION`).

```
<memoryBacking>
  <hugepages/>
</memoryBacking>
```

4. Start your VM Guest and check on the host whether it uses hugepages:

```
>cat /proc/meminfo | grep HugePages_
HugePages_Total:①      512
HugePages_Free:②       92
HugePages_Rsvd:③       0
HugePages_Surp:④       0
```

- ① Size of the pool of huge pages
- ② Number of huge pages in the pool that are not yet allocated
- ③ Number of huge pages for which a commitment to allocate from the pool has been made, but no allocation has yet been made
- ④ Number of huge pages in the pool above the value in `/proc/sys/vm/nr_hugepages`. The maximum number of surplus huge pages is controlled by `/proc/sys/vm/nr_overcommit_hugepages`

4.1.2. Transparent huge pages

Transparent huge pages (THP) provide a way to dynamically allocate huge pages with the `khugepaged` kernel thread, rather than manually managing their allocation and use. Workloads with contiguous memory access patterns can benefit greatly from THP. A 1000 fold decrease in page faults can be observed when running synthetic workloads with contiguous memory access patterns. Conversely, workloads with sparse memory access patterns (like databases) may perform poorly with THP. In such cases, it may be preferable to disable THP by adding the kernel parameter

`transparent_hugepage=never`, rebuild your grub2 configuration, and reboot. Verify if THP is disabled with:

```
>cat /sys/kernel/mm/transparent_hugepage/enabled
always madvise [never]
```

If disabled, the value `never` is shown in square brackets like in the example above.



Xen

THP is not available under Xen.

4.1.3. Xen-specific memory notes

4.1.3.1. Managing domain-0 memory

In previous versions of SUSE Linux Enterprise Server, the default memory allocation scheme of a Xen host was to allocate all host physical memory to Dom0 and enable auto-ballooning. Memory was automatically ballooned from Dom0 when additional domains were started. This behavior has always been error prone and disabling it was strongly encouraged. Starting in SUSE Linux Enterprise Server 15 SP1, auto-ballooning has been disabled by default and Dom0 is given 10% of host physical memory + 1 GB. For example, on a host with 32 GB of physical memory, 4.2 GB of memory is allocated to Dom0.

The use of `dom0_mem` Xen command-line option in `/etc/default/grub` is still supported and encouraged (see *the section called “Change kernel parameters at boot time”* for more information). You can restore the old behavior by setting `dom0_mem` to the host physical memory size and enabling the `autoballoon` setting in `/etc/xen/xl.conf`.

4.1.4. KSM and page sharing

Kernel Samepage Merging is a kernel feature that reduces memory consumption on the VM Host Server by sharing blocks of memory that VM Guests have in common. The KSM daemon `ksmd` periodically scans user memory, looking for pages with identical contents, which can be replaced by a single write-protected page. To enable the KSM service, first make sure that the package `qemu-ksm` is installed, then run the command:

```
>sudo systemctl enable --now ksm.service
```

Alternatively, it can also be started by running the command:

```
# echo 1 > /sys/kernel/mm/ksm/run
```

One advantage of using KSM from a VM Guest's perspective is that all guest memory is backed by host anonymous memory. You can share *pagecache*, *tmpfs* or any kind of memory allocated in the guest.

KSM is controlled by `sysfs`. You can check KSM's values in `/sys/kernel/mm/ksm/`:

- `pages_shared`: the number of shared pages that are being used (read-only).
- `pages_sharing`: the number of sites sharing the pages (read-only).
- `pages_unshared`: the number of pages that are unique and repeatedly checked for merging (read-only).
- `pages_volatile`: the number of pages that are changing too fast to be considered for merging (read-only).
- `full_scans`: the number of times all mergeable areas have been scanned (read-only).
- `sleep_millisecs`: the number of milliseconds `ksmd` should sleep before the next scan. A low value will overuse the CPU, consuming CPU time that could be used for other tasks. We recommend a value greater than 1000.
- `pages_to_scan`: the number of present pages to scan before `ksmd` goes to sleep. A high value will overuse the CPU. We recommend starting with a value of 1000 and then adjusting based on the KSM results observed while testing your deployment.
- `merge_across_nodes`: by default, the system merges pages across NUMA nodes. Set this option to 0 to disable this behavior.



Use cases

KSM is a good technique to over-commit host memory when running multiple instances of the same application or VM Guest. When applications and VM Guest are heterogeneous and do not share any common data, it is preferable to disable KSM. To do that, run:

```
>sudo systemctl disable --now ksm.service
```

Alternatively, it can also be disabled by running the command:

```
# echo 0 > /sys/kernel/mm/ksm/run
```

In a mixed heterogeneous and homogeneous environment, KSM can be enabled on the host but disabled on a per VM Guest basis. Use **virsh edit** to disable page sharing of a VM Guest by adding the following to the guest's XML configuration:

```
<memoryBacking>
  <nosharepages/>
</memoryBacking>
```



Avoid out-of-memory conditions

KSM can free up certain memory on the host system, but the administrator should reserve enough swap to avoid out-of-memory conditions if that shareable memory decreases. If the amount of shareable memory decreases, the use of physical memory is increased.



KSM as a side channel

Because of its nature, KSM can form a side channel between otherwise isolated guests. It is discouraged to enable KSM in environments where guests from different security domains are executed.



Memory access latencies

By default, KSM will merge common pages across NUMA nodes. If the merged, common page is now located on a distant NUMA node (relative to the node running the VM Guest vCPUs), this may degrade VM Guest performance. If increased memory access latencies are noticed in the VM Guest, disable cross-node merging with the `merge_across_nodes` sysfs control:

```
# echo 0 > /sys/kernel/mm/ksm/merge_across_nodes
```

4.1.5. VM Guest: memory hotplug

To optimize the usage of your host memory, it may be useful to hotplug more memory for a running VM Guest when required. To support memory hotplugging, you must first configure the `<maxMemory>` tag in the VM Guest's configuration file:

```
<maxMemory① slots='16'② unit='KiB'>20971520③</maxMemory>
  <memory④ unit='KiB'>1048576</memory>
<currentMemory⑤ unit='KiB'>1048576</currentMemory>
```

- ① Runtime maximum memory allocation of the guest
- ② Number of slots available for adding memory to the guest
- ③ Valid units are:

- “KB” for kilobytes (1,000 bytes)
- “k” or “KiB” for kibibytes (1,024 bytes)
- “MB” for megabytes (1,000,000 bytes)
- “M” or “MiB” for mebibytes (1,048,576 bytes)
- “GB” for gigabytes (1,000,000,000 bytes)
- “G” or “GiB” for gibibytes (1,073,741,824 bytes)
- “TB” for terabytes (1,000,000,000,000 bytes)
- “T” or “TiB” for tebibytes (1,099,511,627,776 bytes)

④ Maximum allocation of memory for the guest at boot time

⑤ Actual allocation of memory for the guest

To hotplug memory devices into the slots, create a file `mem-dev.xml` like the following:

```
<memory model='dimm'>
  <target>
    <size unit='KiB'>524287</size>
    <node>0</node>
  </target>
</memory>
```

And attach it with the following command:

```
>virsh attach-device vm-name mem-dev.xml
```

For memory device hotplug, the guest must have at least 1 NUMA cell defined (see *the section called “VM Guest virtual NUMA topology”*).

4.2. Swap

Swap is used by the system to store underused physical memory (low usage, or not accessed for a long time). To prevent the system from running out of memory, setting up a minimum swap is highly recommended.

4.2.1. swappiness

The `swappiness` setting controls your system's swap behavior. It defines how memory pages are swapped to disk. A high value of *swappiness* results in a system that swaps more often. Available values range from 0 to 200. A value of 200 tells the system to find inactive pages and put them in swap. A value of 0 disables swapping.

To test on a live system, change the value of `/proc/sys/vm/swappiness` on the fly and check the memory usage afterward:


```
# echo 35 > /proc/sys/vm/swappiness
>free -h
total        used        free        shared    buffers        cached
Mem:      24616680    4991492    19625188        167056    144340    2152408
-/+ buffers/cache:    2694744    21921936
Swap:      6171644         0     6171644
```

To permanently set a swappiness value, add a line in `/etc/sysctl.conf`, for example:

```
vm.swappiness = 35
```

You can also control the swap by using the `swap_hard_limit` element in the XML configuration of your VM Guest. Before setting this parameter and using it in a production environment, test it because the host can stop the domain if the value is too low.

```
<memtune>①
  <hard_limit unit='G'>1</hard_limit>②
  <soft_limit unit='M'>128</soft_limit>③
  <swap_hard_limit unit='G'>2</swap_hard_limit>④
</memtune>
```

- ① This element provides memory tunable parameters for the domain. If this is omitted, it defaults to the defaults provided by the operating system.
- ② Maximum memory the guest can use. To avoid any problems on the VM Guest, we strongly recommend not to use this parameter.
- ③ The memory limit to enforce during memory contention.
- ④ The maximum memory plus swap the VM Guest can use.

4.3. I/O

4.3.1. I/O scheduler

The I/O scheduler for SUSE Linux Enterprise 15 SP2 and up is Budget Fair Queueing (BFQ). The main aim of the BFQ scheduler is to provide a fair allocation of the disk I/O bandwidth for all processes that request an I/O operation. You can have different I/O schedulers for different devices.

To get better performance in host and VM Guest, use `none` in the VM Guest (disable the I/O scheduler) and the `mq-deadline` scheduler for a virtualization host.

Procedure 2. Checking and changing the I/O scheduler at runtime

1. To check your current I/O scheduler for your disk (replace `sdX` by the disk you want to check), run:

```
>cat /sys/block/sdX/queue/scheduler  
mq-deadline kyber [bfq] none
```

The value in square brackets is the one currently selected (bfq in the example above).

2. You can change the scheduler at runtime by running the following command as root:

```
# echo mq-deadline > /sys/block/sdX/queue/scheduler
```

If you need to specify different I/O schedulers for each disk, create the file `/usr/lib/tmpfiles.d/IO_ioscheduler.conf` with content similar to the following example. It defines the mq-deadline scheduler for `/dev/sda` and the none scheduler for `/dev/sdb`. Keep in mind that the device name can be different depending on the device type. This feature is available on SLE 12 and up.

```
w /sys/block/sda/queue/scheduler - - - - mq-deadline  
w /sys/block/sdb/queue/scheduler - - - - none
```

4.3.2. Asynchronous I/O

Many of the virtual disk back-ends use Linux Asynchronous I/O (aio) in their implementation. By default, the maximum number of aio contexts is set to 65536, which can be exceeded when running hundreds of VM Guests using virtual disks serviced by Linux Asynchronous I/O. When running large numbers of VM Guests on a VM Host Server, consider increasing `/proc/sys/fs/aio-max-nr`.

Procedure 3. Checking and changing aio-max-nr at runtime

1. To check your current aio-max-nr setting run:

```
>cat /proc/sys/fs/aio-max-nr  
65536
```

2. You can change aio-max-nr at runtime with the following command:

```
# echo 131072 > /proc/sys/fs/aio-max-nr
```

To permanently set `aio-max-nr`, add an entry to a custom sysctl file. For example, include the following to `/etc/sysctl.d/aio-max-nr.conf`:

```
fs.aio-max-nr = 1048576
```

4.3.3. I/O Virtualization

SUSE products support multiple I/O virtualization technologies. The following table lists advantages and disadvantages of each technology. For more information about I/O in virtualization refer to *the section called "I/O virtualization"*.

Table 2. I/O Virtualization solutions

Technology	Advantage	Disadvantage
Device Assignment (pass-through)	Device accessed directly by the guest	No sharing among multiple guests
	High performance	Live migration is complex
		PCI device limit is 8 per guest
		Limited number of slots on a server
Full virtualization (IDE, SATA, SCSI, e1000)	VM Guest compatibility	Bad performance
	Easy for live migration	Emulated operation
Para-virtualization (virtio-blk, virtio-net, virtio-scsi)	Good performance	Modified guest (PV drivers)
	Easy for live migration	
	Efficient host communication with VM Guest	

4.4. Storage and file system

Storage space for VM Guests can either be a block device, for example, a partition on a physical disk, or an image file on the file system:

Table 3. Block devices compared to disk images

Technology	Advantages	Disadvantages
Block devices	• Better performance • Use standard tools for administration/disk modification • Accessible from host (pro and con)	• Device management

Technology	Advantages	Disadvantages
Image files	• Easier system management • Easily move, clone, expand, back up domains • Comprehensive toolkit (guestfs) for image manipulation • Reduce overhead through sparse files • Fully allocate for best performance	• Lower performance than block devices

For detailed information about image formats and maintaining images, refer to *the section called “VM Guest images”*.

If your image is stored on an NFS share, check certain server and client parameters to improve access to the VM Guest image.

4.4.1. NFS read/write (client)

Options `rsize` and `wsizes` specify the size of the chunks of data that the client and server pass back and forth to each other. You should ensure NFS read/write sizes are sufficiently large, especially for large I/O. Change the `rsize` and `wsizes` parameter in your `/etc/fstab` by increasing the value to 16 KB. This will ensure that all operations can be frozen if there is any instance of hanging.

```
nfs_server:/exported/vm_images① /mnt/images② nfs③ rw④,hard⑤, sync⑥, rsize=8192⑦, wsize=
```

① NFS server's host name and export path.

② Where to mount the NFS exported share.

③ This is an `nfs` mount point.

④ This mount point will be accessible in read/write.

⑤ Determines the recovery behavior of the NFS client after an NFS request times out. `hard` is the best option to avoid data corruption.

⑥ Any system call that writes data to files on that mount point causes that data to be flushed to the server before the system call returns control to user space.

- ⑦ Maximum number of bytes in each network READ request that the NFS client can receive when reading data from a file on an NFS server.
- ⑧ Maximum number of bytes per network WRITE request that the NFS client can send when writing data to a file on an NFS server.

4.4.2. NFS threads (server)

Your NFS server should have enough NFS threads to handle multi-threaded workloads. Use the **nfsstat** tool to get RPC statistics on your server:

```
>sudo nfsstat -rc
Client rpc stats:
calls      retrans    authrefrsh
6401066     198         0           0
```

If the `retrans` is equal to 0, everything is fine. Otherwise, the client needs to retransmit, so increase the `USE_KERNEL_NFSD_NUMBER` variable in `/etc/sysconfig/nfs`, and adjust accordingly until `retrans` is equal to 0.

4.5. CPUs

Host CPU “components” will be “translated” to virtual CPUs in a VM Guest when being assigned. These components can either be:

- *CPU processor*: this describes the main CPU unit, which normally has multiple cores and may support Hyper-Threading.
- *CPU core*: a main CPU unit can provide more than one core, and the proximity of cores speeds up the computation process and reduces energy costs.
- *CPU Hyper-Threading*: this implementation is used to improve the parallelization of computations, but this is not as efficient as a dedicated core.

4.5.1. Assigning CPUs

CPU overcommit occurs when the cumulative number of virtual CPUs of all VM Guests becomes higher than the number of host CPUs. Best performance is achieved when there is no overcommit and each virtual CPU matches one hardware processor or core on the VM Host Server. VM Guests running on an overcommitted host will experience increased latency and a negative effect on per-VM Guest throughput is often observed. Therefore, try to avoid overcommitting CPUs.

Deciding whether to allow CPU overcommit or not requires good a priori knowledge of workload as a whole. For example, if you know that all the VM Guests' virtual CPUs will not be loaded over 50%, then you can assume that overcommitting the host by a factor of 2 (which means having 128

virtual CPUs in total, on a host with 64 CPUs) will work well. However, if you know that all the virtual CPUs of the VM Guests will try to run at 100% for most of the time, then even having one virtual CPU more than the host has CPUs is already a misconfiguration.

Overcommitting to a point where the cumulative number of virtual CPUs is higher than 8 times the number of physical cores of the VM Host Server may lead to a malfunctioning and unstable system and should hence be avoided.

Unless you know exactly how many virtual CPUs are required for a VM Guest, start with one. Target a CPU workload of approximately 70% inside your VM (see *the section called “Processes”* for information on monitoring tools). If you allocate more processors than needed in the VM Guest, this will negatively affect the performance of host and guest. Cycle efficiency will be degraded, as the unused vCPU will still cause timer interrupts. In case you primarily run single threaded applications on a VM Guest, a single virtual CPU is the best choice.

A single VM Guest with more virtual CPUs than the VM Host Server has CPUs is always a misconfiguration.

4.5.2. VM Guest CPU configuration

This section describes how to choose and configure a CPU type for a VM Guest. You will also learn how to pin virtual CPUs to physical CPUs on the host system. For more information about virtual CPU configuration and tuning parameters, refer to the libvirt documentation at <https://libvirt.org/formatdomain.html#elementsCPU>.

4.5.2.1. Virtual CPU models and features

The CPU model and topology can be specified individually for each VM Guest. Configuration options range from selecting specific CPU models to excluding certain CPU features. Predefined CPU models are listed in files in the directory `/usr/share/libvirt/cpu_map/`. A CPU model and topology that is similar to the host generally provides the best performance. The host system CPU model and topology can be displayed by running `virsh capabilities`.

Changing the default virtual CPU configuration will require a VM Guest shutdown when migrating it to a host with different hardware. More information on VM Guest migration is available in *Chapter 17, Migrating VM Guests*.

To specify a particular CPU model for a VM Guest, add a respective entry to the VM Guest configuration file. The following example configures a Broadwell CPU with the invariant TSC feature:

```
<cpu mode='custom' match='exact'>
  <model>Broadwell</model>
  <feature name='invttsc' />
</cpu>
```

For a virtual CPU that most closely resembles the host physical CPU, `<cpu mode='host-pass-through'>` can be used. A `host-passthrough` CPU model may not exactly resemble the host physical CPU, since, by default, KVM will mask any non-migratable features. For example, `invttsc` is not included in the virtual CPU feature set. Changing the default KVM behavior is not directly supported through `libvirt`, although it does allow arbitrary pass-through of KVM command-line arguments. Continuing with the `invttsc` example, you can achieve pass-through of the host CPU (including `invttsc`) with the following command-line pass-through in the VM Guest configuration file:

```
<domain type='kvm' xmlns:qemu='http://libvirt.org/schemas/domain/qemu/1.0'>
  <qemu:commandline>
    <qemu:arg value='-cpu' />
    <qemu:arg value='host,migratable=off,+invttsc' />
  </qemu:commandline>
  ...
</domain>
```



The host-passthrough mode

Since `host-passthrough` exposes the physical CPU details to the virtual CPU, migration to dissimilar hardware is not possible. See *the section called “Virtual CPU migration considerations”* for more information.

4.5.2.2. Virtual CPU pinning

Virtual CPU pinning is used to constrain virtual CPU threads to a set of physical CPUs. The `vcupin` element specifies the physical host CPUs that a virtual CPU can use. If this element is not set and the attribute `cpuset` of the `vcpu` element is not specified, the virtual CPU is free to use any of the physical CPUs.

CPU intensive workloads can benefit from virtual CPU pinning by increasing the physical CPU cache hit ratio. To pin a virtual CPU to a specific physical CPU, run the following commands:

```
>virsh vcpupin DOMAIN_ID --vcpu VCPU_NUMBER
VCPU: CPU Affinity
-----
0: 0-7
#virsh vcpupin SLE15 --vcpu 0 0 --config
```

The last command generates the following entry in the XML configuration:

```
<cpupin>
  <vcpupin vcpu='0' cpuset='0' />
</cpupin>
```



Virtual CPU pinning on NUMA nodes

To confine a VM Guest's CPUs and its memory to a NUMA node, you can use virtual CPU pinning and memory allocation policies on a NUMA system. See *the section called “NUMA tuning”* for more information related to NUMA tuning.



Virtual CPU pinning and live migration

Even though `vcpupin` can improve performance, it can complicate live migration. See *the section called “Virtual CPU migration considerations”* for more information on virtual CPU migration considerations.

4.5.2.3. Virtual CPU migration considerations

Selecting a virtual CPU model containing all the latest features may improve performance of a VM Guest workload, but often at the expense of migratability. Unless all hosts in the cluster contain the latest CPU features, migration can fail when a destination host lacks the new features. If migratability of a virtual CPU is preferred over the latest CPU features, a normalized CPU model and feature set should be used. The `virsh cpu-baseline` command can help define a normalized virtual CPU that can be migrated across all hosts. The following command, when run on each host in the migration cluster, illustrates the collection of all hosts' capabilities in `all-hosts-caps.xml`.

```
>sudo virsh capabilities >> all-hosts-cpu-caps.xml
```

With the capabilities of each host collected in `all-hosts-caps.xml`, use `virsh cpu-baseline` to create a virtual CPU definition that will be compatible across all hosts.

```
>sudo virsh cpu-baseline all-hosts-caps.xml
```

The resulting virtual CPU definition can be used as the `cpu` element in the VM Guest configuration file.

At a logical level, virtual CPU pinning is a form of hardware pass-through. CPU pinning couples physical resources to virtual resources, which can also be problematic for migration. For example, the migration will fail if the requested physical resources are not available on the destination host, or if the source and destination hosts have different NUMA topologies. For more recommendations about Live Migration, see *the section called “Migration requirements”*.

4.6. NUMA tuning

NUMA is an acronym for Non Uniform Memory Access. A NUMA system has multiple physical CPUs, each with local memory attached. Each CPU can also access other CPUs' memory, known as “remote memory access”, but it is much slower than accessing local memory. NUMA systems can negatively affect VM Guest performance if not tuned properly. Although ultimately tuning is workload dependent, this section describes controls that should be considered when deploying VM Guests on NUMA hosts. Always consider your host topology when configuring and deploying VMs.

SUSE Linux Enterprise Server contains a NUMA auto-balancer that strives to reduce remote memory access by placing memory on the same NUMA node as the CPU processing it. Standard tools such as **cgset** and virtualization tools such as libvirt provide mechanisms to constrain VM Guest resources to physical resources.

numactl is used to check for host NUMA capabilities:

```
>sudo numactl --hardware
available: 4 nodes (0-3)
node 0 cpus: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 72 73 74 75 76 77 78
79 80 81 82 83 84 85 86 87 88 89
node 0 size: 31975 MB
node 0 free: 31120 MB
node 1 cpus: 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 90 91 92 93
94 95 96 97 98 99 100 101 102 103 104 105 106 107
node 1 size: 32316 MB
node 1 free: 31673 MB
node 2 cpus: 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 108 109 110
111 112 113 114 115 116 117 118 119 120 121 122 123 124 125
node 2 size: 32316 MB
node 2 free: 31726 MB
node 3 cpus: 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 126 127 128
129 130 131 132 133 134 135 136 137 138 139 140 141 142 143
node 3 size: 32314 MB
node 3 free: 31387 MB
node distances:
node  0  1  2  3
0:  10  21  21  21
1:  21  10  21  21
2:  21  21  10  21
3:  21  21  21  10
```

The **numactl** output shows this is a NUMA system with 4 nodes or cells, each containing 36 CPUs and approximately 32G memory. **virsh capabilities** can also be used to examine the systems NUMA capabilities and CPU topology.

4.6.1. NUMA balancing

On NUMA machines, there is a performance penalty if remote memory is accessed by a CPU. Automatic NUMA balancing scans a task's address space and unmaps pages. By doing so, it

detects whether pages are properly placed or whether to migrate the data to a memory node local to where the task is running. In defined intervals (configured with `numa_balancing_scan_delay_ms`), the task scans the next scan size number of pages (configured with `numa_balancing_scan_size_mb`) in its address space. When the end of the address space is reached, the scanner restarts from the beginning.

Higher scan rates cause higher system overhead as page faults must be trapped and data needs to be migrated. However, the higher the scan rate, the more quickly a task's memory migrates to a local node when the workload pattern changes. This minimizes the performance impact caused by remote memory accesses. These `sysctl` directives control the thresholds for scan delays and the number of pages scanned:

```
>sudo sysctl -a | grep numa_balancing
kernel.numa_balancing = 1❶
kernel.numa_balancing_scan_delay_ms = 1000❷
kernel.numa_balancing_scan_period_max_ms = 60000❸
kernel.numa_balancing_scan_period_min_ms = 1000❹
kernel.numa_balancing_scan_size_mb = 256❺
```

- ❶ Enables/disables automatic page fault-based NUMA balancing
- ❷ Starting scan delay used for a task when it initially forks
- ❸ Maximum time in milliseconds to scan a task's virtual memory
- ❹ Minimum time in milliseconds to scan a task's virtual memory
- ❺ Size in megabytes' worth of pages to be scanned for a given scan

For more information, see *Chapter 11, Automatic Non-Uniform Memory Access (NUMA) balancing*.

The main goal of automatic NUMA balancing is either to reschedule tasks on the same node's memory (so the CPU follows the memory), or to copy the memory's pages to the same node (so the memory follows the CPU).



Task placement

There are no rules to define the best place to run a task, because tasks could share memory with other tasks. For the best performance, we recommend to group tasks sharing memory on the same node. Check NUMA statistics with `# cat /proc/vmstat | grep numa_`.

4.6.2. Memory allocation control with the Cpuset controller

The cgroups cpuset controller can be used to confine memory used by a process to a NUMA node. There are three cpuset memory policy modes available:

- `interleave`: this is a memory placement policy which is also known as round-robin. This policy can provide substantial improvements for jobs that need to place thread local data on the corresponding node. When the interleave destination is not available, it will be moved to another node.
- `bind`: this will place memory only on one node, which means in case of insufficient memory, the allocation will fail.
- `preferred`: this policy will apply a preference to allocate memory to a node. If there is not enough space for memory on this node, it will fall back to another node.

You can change the memory policy mode with the `cgset` tool from the `libcgroup-tools` package:

```
>sudo cgset -r cpuset.mems=NODE sysdefault/libvirt/qemu/KVM_NAME/emulator
```

To migrate pages to a node, use the `migratepages` tool:

```
>migratepages PIDFROM-NODETO-NODE
```

To check everything is fine, use: `cat /proc/PID/status | grep Cpus`.



Kernel NUMA/cpuset memory policy

For more information, see [Kernel NUMA memory policy](#) and [cpusets memory policy](#). Check also the [Libvirt NUMA Tuning documentation](#).

4.6.3. VM Guest: NUMA related configuration

`libvirt` allows to set up virtual NUMA and memory access policies. Configuring these settings is not supported by `virt-install` or `virt-manager` and needs to be done manually by editing the VM Guest configuration file with `virsh edit`.

4.6.3.1. VM Guest virtual NUMA topology

Creating a VM Guest virtual NUMA (vNUMA) policy that resembles the host NUMA topology can often increase performance of traditional large, scale-up workloads. VM Guest vNUMA topology can be specified using the `numa` element in the XML configuration:

```
<cpu>
...
  <numa>
    <cell❶ id="0"❷ cpus='0-1'❸ memory='512000' unit='KiB' />
    <cell id="1" cpus='2-3' memory='256000'❹
      unit='KiB'❺ memAccess='shared'❻ />
  </numa>
...
</cpu>
```

- ❶ Each `cell` element specifies a vNUMA cell or node
- ❷ All cells should have an `id` attribute, allowing to reference the cell in other configuration blocks. Otherwise, cells are assigned IDs in ascending order starting from 0.
- ❸ The CPU or range of CPUs that are part of the node
- ❹ The node memory
- ❺ Units in which node memory is specified
- ❻ Optional attribute which can control whether the memory is to be mapped as `shared` or `private`. This is valid only for hugepages-backed memory.

To find where the VM Guest has allocated its pages. use: `cat /proc/PID/numa_maps` and `cat /sys/fs/cgroup/memory/sysdefault/libvirt/qemu/KVM_NAME/memory.numa_stat`.



NUMA specification

The `libvirt` VM Guest NUMA specification is currently only available for QEMU/KVM.

4.6.3.2. Memory allocation control with `libvirt`

If the VM Guest has a vNUMA topology (see *the section called “VM Guest virtual NUMA topology”*), memory can be pinned to host NUMA nodes using the `numatune` element. This method is currently only available for QEMU/KVM guests. See *Non-vNUMA VM Guest* for how to configure non-vNUMA VM Guests.

```
<numatune>
  <memory mode="strict"❶ nodeset="1-4,^3"❷/>
  <memnode❸ cellid="0"❹ mode="strict" nodeset="1"/>
  <memnode cellid="2" placement="strict"❺ mode="preferred" nodeset="2"/>
</numatune>
```

- ❶ Policies available are: `interleave` (round-robin like), `strict` (default) or `preferred`.
- ❷ Specify the NUMA nodes.
- ❸ Specify memory allocation policies for each guest NUMA node (if this element is not defined, then this will fall back and use the `memory` element).
- ❹ Addresses the guest NUMA node for which the settings are applied.
- ❺ The placement attribute can be used to indicate the memory placement mode for a domain process, the value can be `auto` or `strict`.

Non-vNUMA VM Guest



On a non-vNUMA VM Guest, pinning memory to host NUMA nodes is done as in the following example:

```
<numatune>
  <memory mode="strict" nodeset="0-1"/>
</numatune>
```

In this example, memory is allocated from the host nodes 0 and 1. If these memory requirements cannot be fulfilled, starting the VM Guest will fail. `virt-install` also supports this configuration with the `--numatune` option.

Memory and CPU across NUMA nodes



You should avoid allocating VM Guest memory across NUMA nodes, and prevent virtual CPUs from floating across NUMA nodes.

5. VM Guest images

Images are virtual disks used to store the operating system and data of VM Guests. They can be created, maintained and queried with the `qemu-img` command. Refer to *the section called “Creating, converting, and checking disk images”* for more information on the `qemu-img` tool and examples.

5.1. VM Guest image formats

Certain storage formats which QEMU recognizes have their origins in other virtualization technologies. By recognizing these formats, QEMU can use either data stores or entire guests that were originally targeted to run under these other virtualization technologies. Certain formats are supported only in read-only mode. To use them in read/write mode, convert them to a fully supported QEMU storage format (using `qemu-img`). Otherwise they can only be used as read-only data store in a QEMU guest.

Use `qemu-img info VMGUEST.IMG` to get information about an existing image, such as the format, the virtual size, the physical size and snapshots, if available.



Performance

It is recommended to convert the disk images to either raw or qcow2 to achieve good performance.



Encrypted images cannot be compressed

When you create an image, you cannot use compression (`-c`) in the output file together with the encryption option (`-e`).

5.1.1. Raw format

- This format is simple and easily exportable to all other emulators/hypervisors.
- It provides the best performance (least I/O overhead).
- It occupies all allocated space on the file system.
- The raw format allows to copy a VM Guest image to a physical device (`dd if=VMGUEST.RAW of=/dev/sda`).
- It is byte-for-byte the same as what the VM Guest sees, so this wastes a lot of space.

5.1.2. qcow2 format

- Use this to have smaller images (useful if your file system does not support holes).
- It has optional AES encryption (now deprecated).
- Zlib-based compression option.
- Supports multiple VM snapshots (internal, external).

- Improved performance and stability.
- Supports changing the backing file.
- Supports consistency checks.
- Less performance than raw format.

l2-cache-size

qcow2 can provide the same performance for random read/write access as raw format, but it needs a well-sized cache size. By default, cache size is set to 1 MB. This will give good performance up to a disk size of 8 GB. If you need a bigger disk size, you need to adjust the cache size. For a disk size of 64 GB ($64 \times 1024 = 65536$), you need $65536 / 8192 \text{ B} = 8 \text{ MB}$ of cache (`-drive format=qcow2,l2-cache-size=8M`).

Cluster size

The qcow2 format offers the capability to change the cluster size. The value must be between 512 KB and 2 MB. Smaller cluster sizes can improve the image file size, whereas larger cluster sizes provide better performance.

Preallocation

An image with preallocated metadata is initially larger but can improve performance when the image needs to grow.

Lazy refcounts

Reference count updates are postponed with the goal of avoiding metadata I/O and improving performance. This is particularly beneficial with `cache=writethrough`. This option does not batch metadata updates, but in case of host crash, the reference count tables must be rebuilt, this is done automatically at the next open with `qemu-img check -r all` and takes a certain amount of time.

5.1.3. qed format

qed is a follow-on qcow (QEMU Copy On Write) format. Because qcow2 provides all the benefits of qed and more, qed is now deprecated.

5.1.4. VMDK format

VMware 3, 4 or 6 image format for exchanging images with that product.

5.2. Overlay disk images

The qcow2 and qed formats provide a way to create a base image (also called backing file) and overlay images on top of the base image. A backing file is useful for reverting to a known state and discarding the overlay. If you write to the image, the backing image will be untouched and all changes will be recorded in the overlay image file. The backing file will never be modified unless you use the `commit` monitor command (or `qemu-img commit`).

To create an overlay image:

```
#qemu-img create -o①backing_file=vmquest.raw②,backing_fmt=raw③\  
-f④ qcow2 vmquest.cow⑤
```

- ① Use `-o ?` for an overview of available options.
- ② The backing file name.
- ③ Specify the file format for the backing file.
- ④ Specify the image format for the VM Guest.
- ⑤ Image name of the VM Guest, it will only record the differences from the backing file.



Backing image path

You should not change the path to the backing image, otherwise you will need to adjust it. The path is stored in the overlay image file. To update the path, make a symbolic link from the original path to the new path and then use the `qemu-img rebase` option.

```
#ln -sf /var/lib/images/OLD_PATH/vmquest.raw \  
/var/lib/images/NEW_PATH/vmquest.raw  
#qemu-img rebase① -u② -b③ \  
/var/lib/images/OLD_PATH/vmquest.raw /var/lib/images/NEW_PATH/vmquest.cow
```

- ① The `rebase` subcommand tells `qemu-img` to change the backing file image.
- ② The `-u` option activates the unsafe mode (see note below). There are two different modes in which `rebase` can operate:
 - **Safe**: this is the default mode and performs a real rebase operation. The safe mode is a time-consuming operation.
 - **Unsafe**: the unsafe mode (`-u`) only changes the backing files name and the format of the file name, making no checks on the file's contents. Use this mode to rename or move a backing file.

- ③ The backing image to be used is specified with `-b` and the image path is the last argument of the command.

A common use is to initiate a new guest with the backing file. Let's assume we have a `sle15_base.img` VM Guest ready to be used (fresh installation without any modification). This will be our backing file. Now you need to test a new package, on an updated system and on a system with a different kernel. We can use `sle15_base.img` to instantiate the new SUSE Linux Enterprise VM Guest by creating a qcow2 overlay file pointing to this backing file (`sle15_base.img`).

In our example, we will use `sle15_updated.qcow2` for the updated system, and `sle15_kernel.qcow2` for the system with a different kernel.

To create the two thin provisioned systems, use the `qemu-img` command line with the `-b` option:

```
#qemu-img create -b /var/lib/libvirt/sle15_base.img -f qcow2 \
/var/lib/libvirt/sle15_updated.qcow2
Formatting 'sle15_updated.qcow2', fmt=qcow2 size=17179869184
backing_file='sle15_base.img' encryption=off cluster_size=65536
lazy_refcounts=off nocow=off
#qemu-img create -b /var/lib/libvirt/sle15_base.img -f qcow2 \
/var/lib/libvirt/sle15_kernel.qcow2
Formatting 'sle15_kernel.qcow2', fmt=qcow2 size=17179869184
backing_file='vmguest-sle15_base.img' encryption=off cluster_size=65536
lazy_refcounts=off nocow=off
```

The images are now usable, and you can do your test without touching the initial `sle15_base.img` backing file. All changes will be stored in the new overlay images. You can also use these new images as a backing file and create a new overlay.

```
#qemu-img create -b sle15_kernel.qcow2 -f qcow2 sle15_kernel_TEST.qcow2
```

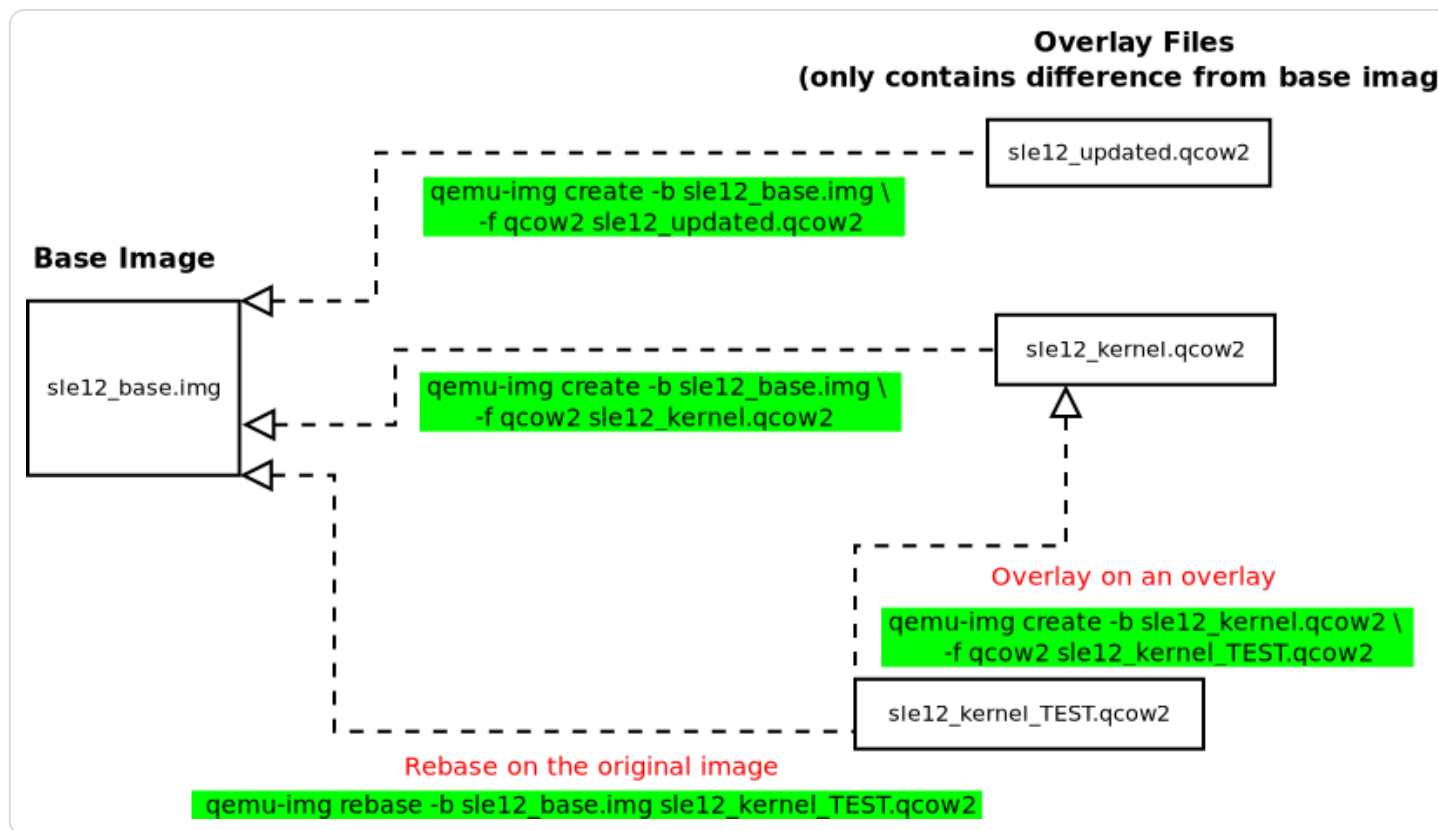
When using `qemu-img info` with the option `--backing-chain`, it will return all information about the entire backing chain recursively:

```
#qemu-img info --backing-chain
/var/lib/libvirt/images/sle15_kernel_TEST.qcow2
image: sle15_kernel_TEST.qcow2
file format: qcow2
virtual size: 16G (17179869184 bytes)
disk size: 196K
cluster_size: 65536
backing_file: sle15_kernel.qcow2
Format specific information:
compat: 1.1
lazy refcounts: false

image: sle15_kernel.qcow2
file format: qcow2
virtual size: 16G (17179869184 bytes)
disk size: 196K
cluster_size: 65536
backing_file: SLE15.qcow2
Format specific information:
compat: 1.1
lazy refcounts: false

image: sle15_base.img
file format: qcow2
virtual size: 16G (17179869184 bytes)
disk size: 16G
cluster_size: 65536
Format specific information:
compat: 1.1
lazy refcounts: true
```

Figure 1. Understanding image overlay



5.3. Opening a VM Guest image

To access the file system of an image, use the `guestfs-tools`. If you do not have this tool installed on your system, you can mount an image with other Linux tools. Avoid accessing an untrusted or unknown VM Guest's image system because this can lead to security issues (for more information, read [D. Berrangé's post](#)).

5.3.1. Opening a raw image

Procedure 4. Mounting a raw image

1. To mount the image, find a free loop device. The following command displays the first unused loop device, `/dev/loop1` in this example.

```
#losetup -f  
/dev/loop1
```

2. Associate an image (`SLE15.raw` in this example) with the loop device:

```
#losetup /dev/loop1 SLE15.raw
```

3. Check whether the image has successfully been associated with the loop device by getting detailed information about the loop device:

```
#losetup -l  
NAME                SIZE LIMIT  OFFSET  AUTOCLEAR  RO  BACK-FILE  
/dev/loop1          0          0          0    0    /var/lib/libvirt/images/SLE15.raw
```

4. Check the image's partitions with `kpartx`:

```
#kpartx -a❶ -v❷ /dev/loop1  
add map loop1p1 (254:1): 0 29358080 linear /dev/loop1 2048
```

❶ Add partition device mappings.

❷ Verbose mode.

5. Now mount the image partitions (to `/mnt/sle15mount` in the following example):

```
#mkdir /mnt/sle15mount  
#mount /dev/mapper/loop1p1 /mnt/sle15mount
```



Raw image with LVM

If your raw image contains an LVM volume group, you should use LVM tools to mount the partition. Refer to *the section called “Opening images containing LVM”*.

Procedure 5. Unmounting a raw image

1. Unmount all mounted partitions of the image, for example:

```
#umount /mnt/sle15mount
```

2. Delete partition device mappings with `kpartx`:

```
#kpartx -d /dev/loop1
```

3. Detach the devices with **losetup**

```
#losetup -d /dev/loop1
```

5.3.2. Opening a qcow2 image

Procedure 6. Mounting a qcow2 image

1. First, you need to load the `nbdk` (network block devices) module. The following example loads it with support for 16 block devices (`max_part=16`). Check with **dmesg** whether the operation was successful:

```
#modprobe nbdk max_part=16
#dmesg | grep nbdk
[89155.142425] nbdk: registered device at major 43
```

2. Connect the VM Guest image, for example, `SLE15.qcow2`, to an NBD device (`/dev/nbd0` in the following example) with the **qemu-nbd** command. Use a free NBD device:

```
#qemu-nbd -c❶ /dev/nbd0❷ SLE15.qcow2❸
```

- ❶ Connect `SLE15.qcow2` to the local NBD device `/dev/nbd0`
- ❷ NBD device to use
- ❸ VM Guest image to use



Checking for a free NBD device

To check whether an NBD device is free, run the following command:

```
#lsof /dev/nbd0
COMMAND    PID USER   FD   TYPE DEVICE SIZE/OFF  NODE NAME
qemu-nbd  15149 root    10u   BLK   43,0      0t0 47347 /dev/nbd0
```

If the command produces an output like in the example above, the device is busy (not free). This can also be confirmed by the presence of the `/sys/devices/virtual/block/nbd0/pid` file.

3. Inform the operating system about partition table changes with **partprobe**:

```
#partprobe /dev/nbd0 -s
/dev/nbd0: msdos partitions 1 2
#dmesg | grep nbd0 | tail -1
[89699.082206] nbd0: p1 p2
```

4. In the example above, the `SLE15.qcow2` contains two partitions: `/dev/nbd0p1` and `/dev/nbd0p2`. Before mounting these partitions, use **vgscan** to check whether they belong to an LVM volume:

```
#vgscan -v
Wiping cache of LVM-capable devices
Wiping internal VG cache
Reading all physical volumes. This may take a while...
Using volume group(s) on command line.
No volume groups found.
```

5. If no LVM volume has been found, you can mount the partition with **mount**:

```
#mkdir /mnt/nbd0p2
# mount /dev/nbd0p1 /mnt/nbd0p2
```

Refer to *the section called “Opening images containing LVM”* for information on how to handle LVM volumes.

Procedure 7. Unmounting a qcow2 image

1. Unmount all mounted partitions of the image, for example:

```
#umount /mnt/nbd0p2
```

2. Disconnect the image from the `/dev/nbd0` device.

```
#qemu-nbd -d /dev/nbd0
```

5.3.3. Opening images containing LVM



Warning

If your VM Host Server uses the VG name `system`, and the guest image also uses the VG name `system`, LVM will complain during its activation. A workaround is to temporarily rename the guest VG, while a correct approach is to use different VG names for the guests than for the VM Host Server.

Procedure 8. Mounting images containing LVM

1. To check images for LVM groups, use **vgscan -v**. If an image contains LVM groups, the output of the command looks like the following:

```
#vgscan -v
Wiping cache of LVM-capable devices
Wiping internal VG cache
Reading all physical volumes. This may take a while...
Finding all volume groups
Finding volume group "system"
Found volume group "system" using metadata type lvm2
```

2. The `system` LVM volume group has been found on the system. You can get more information about this volume with **vgdisplay VOLUMEGROUPNAME** (in our case `VOLUMEGROUPNAME` is `system`). You should activate this volume group to expose LVM partitions as devices so the system can mount them. Use **vgchange**:

```
#vgchange -ay -v
Finding all volume groups
Finding volume group "system"
Found volume group "system"
activation/volume_list configuration setting not defined: Checking only
host tags for system/home
Creating system-home
Loading system-home table (254:0)
Resuming system-home (254:0)
Found volume group "system"
activation/volume_list configuration setting not defined: Checking only
host tags for system/root
Creating system-root
Loading system-root table (254:1)
Resuming system-root (254:1)
Found volume group "system"
activation/volume_list configuration setting not defined: Checking only
host tags for system/swap
Creating system-swap
Loading system-swap table (254:2)
Resuming system-swap (254:2)
Activated 3 logical volumes in volume group system
  3 logical volume(s) in volume group "system" now active
```

3. All partitions in the volume group will be listed in the `/dev/mapper` directory. You can simply mount them now.

```
#ls /dev/mapper/system-*
/dev/mapper/system-home /dev/mapper/system-root /dev/mapper/system-swap

#mkdir /mnt/system-root
#mount /dev/mapper/system-root /mnt/system-root

#ls /mnt/system-root/
bin  dev  home  lib64      mnt  proc      root  sbin      srv  tmp  var
boot etc  lib   lost+found opt  read-write run  selinux  sys  usr
```

Procedure 9. Unmounting images containing LVM

1. Unmount all partitions (with `umount`)

```
#umount /mnt/system-root
```

2. Deactivate the LVM volume group (with `vgchange -an VOLUMEGROUPNAME`)

```
#vgchange -an -v system
Using volume group(s) on command line
Finding volume group "system"
Found volume group "system"
Removing system-home (254:0)
Found volume group "system"
Removing system-root (254:1)
Found volume group "system"
Removing system-swap (254:2)
Deactivated 3 logical volumes in volume group system
0 logical volume(s) in volume group "system" now active
```

3. Now you have two choices:

- With a qcow2 image, proceed as described in *step 2* (`qemu-nbd -d /dev/nbd0`).
- With a raw image, proceed as described in *step 2* (`kpartx -d /dev/loop1; losetup -d /dev/loop1`).

Check for a successful unmount



You should double-check that unmounting succeeded by using a system command like `losetup`, `qemu-nbd`, `mount` or `vgscan`. If this is not the case, you may have trouble using the VM Guest because its system image is used in different places.

5.4. File system sharing

You can access a host directory in the VM Guest using the `filesystem` element. In the following example we will share the `/data/shared` directory and mount it in the VM Guest. The `accessmode` parameter only works with `type='mount'` for the QEMU/KVM driver (most other values for `type` are exclusively used for the LXC driver).

```
<filesystem type='mount'❶ accessmode='mapped'❷>
  <source dir='/data/shared'❸>
  <target dir='shared'❹/>
</filesystem>
```

❶ A host directory to mount VM Guest.

❷ Access mode (the security mode) set to `mapped` will give access with the permissions of the hypervisor. Use `passthrough` to access this share with the permissions of the user inside the VM Guest.

❸ Path to share with the VM Guest.

❹ Name or label of the path for the mount command.

To mount the `shared` directory on the VM Guest, use the following commands: under the VM Guest, now you need to mount the `target dir='shared'`:

```
#mkdir /opt/mnt_shared
#mount shared -t 9p /opt/mnt_shared -o trans=virtio
```

See [libvirt File System](#) and [QEMU 9psetup](#) for more information.

6. VM Guest configuration

6.1. Virtio driver

To increase VM Guest performance, we recommend using paravirtualized drivers within the VM Guests. The virtualization standard for such drivers for KVM are the `virtio` drivers, which are designed for running in a virtual environment. Xen uses similar paravirtualized device drivers (like [VMDP](#) in a Windows* guest).

6.1.1. virtio blk

`virtio_blk` is the virtio block device for disk. To use the `virtio blk` driver for a block device, specify the `bus='virtio'` attribute in the disk definition:

```
<disk type='....' device='disk'>
  ....
  <target dev='vda' bus='virtio' />
</disk>
```

Disk device names



`virtio` disk devices are named `/dev/vd[a-z][1-9]`. If you migrate a Linux guest from a non-virtio disk, you need to adjust the `root=` parameter in the GRUB configuration, and regenerate the `initrd` file. Otherwise the system cannot boot. On VM Guests with other operating systems, the boot loader may need to be adjusted or reinstalled accordingly, too.

Using virtio disks with `qemu-system-ARCH`



When running `qemu-system-ARCH`, use the `-drive` option to add a disk to the VM Guest. See *the section called “Basic installation with `qemu-system-ARCH`”* for an example. The `-hd[abcd]` option will not work for virtio disks.

6.1.2. virtio net

`virtio_net` is the virtio network device. The kernel modules should be loaded automatically in the guest at boot time. You need to start the service to make the network available.

```
<interface type='network'>
  ...
  <model type='virtio' />
</interface>
```

6.1.3. virtio balloon

The virtio balloon is used for host memory over-commits for guests. For Linux guests, the balloon driver runs in the guest kernel, whereas for Windows guests, the balloon driver is in the VMDP package. `virtio_balloon` is a PV driver to give or take memory from a VM Guest.

- *Inflate balloon*: return memory from guest to host kernel (for KVM) or to hypervisor (for Xen)
- *Deflate balloon*: Guest will have more available memory

It is controlled by the `currentMemory` and `memory` options.

```
<memory unit='KiB'>16777216</memory>
  <currentMemory unit='KiB'>1048576</currentMemory>
  [...]
  <devices>
    <memballoon model='virtio' />
  </devices>
```

You can also use `virsh` to change it:

```
>virsh setmem DOMAIN_IDMEMORY in KB
```

6.1.4. Checking virtio presence

You can check the virtio block PCI with:

```
>find /sys/devices/ -name virtio*
/sys/devices/pci0000:00/0000:00:06.0/virtio0
/sys/devices/pci0000:00/0000:00:07.0/virtio1
/sys/devices/pci0000:00/0000:00:08.0/virtio2
```

To find the block device associated with `vdX`:

```
>find /sys/devices/ -name virtio* -print -exec ls {}/block 2>/dev/null \;
/sys/devices/pci0000:00/0000:00:06.0/virtio0
/sys/devices/pci0000:00/0000:00:07.0/virtio1
/sys/devices/pci0000:00/0000:00:08.0/virtio2
vda
```

To get more information on the virtio block:

```
>udevadm info -p /sys/devices/pci0000:00/0000:00:08.0/virtio2
P: /devices/pci0000:00/0000:00:08.0/virtio2
E: DEVPATH=/devices/pci0000:00/0000:00:08.0/virtio2
E: DRIVER=virtio_blk
E: MODALIAS=virtio:d00000002v00001AF4
E: SUBSYSTEM=virtio
```

To check all virtio drivers being used:

```
>find /sys/devices/ -name virtio* -print -exec ls -l {}/driver 2>/dev/null \;
/sys/devices/pci0000:00/0000:00:06.0/virtio0
lrwxrwxrwx 1 root root 0 Jun 17 15:48 /sys/devices/pci0000:00/0000:00:06.0/virtio0/driver
/sys/devices/pci0000:00/0000:00:07.0/virtio1
lrwxrwxrwx 1 root root 0 Jun 17 15:47 /sys/devices/pci0000:00/0000:00:07.0/virtio1/driver
/sys/devices/pci0000:00/0000:00:08.0/virtio2
lrwxrwxrwx 1 root root 0 Jun 17 14:35 /sys/devices/pci0000:00/0000:00:08.0/virtio2/driver
```

6.1.5. Find device driver options

Virtio devices and other drivers have multiple options. To list them all, use the `help` parameter of the `qemu-system-ARCH` command.

```
>qemu-system-x86_64 -device virtio-net,help
virtio-net-pci.ioeventfd=on/off
virtio-net-pci.vectors=uint32
virtio-net-pci.indirect_desc=on/off
virtio-net-pci.event_idx=on/off
virtio-net-pci.any_layout=on/off
.....
```

6.2. Cirrus video driver

To get 16-bit color, high compatibility and better performance, we recommend using the `cirrus` video driver.



libvirt

libvirt ignores the `vram` value because video size has been hardcoded in QEMU.

```
<video>
  <model type='cirrus' vram='9216' heads='1' />
</video>
```

6.3. Better entropy

Virtio RNG (random number generator) is a paravirtualized device that is exposed as a hardware RNG device to the guest. On the host side, it can be wired up to one of several sources of entropy (including a real hardware RNG device and the host's `/dev/random`) if hardware support does not exist. The Linux kernel contains the guest driver for the device from version 2.6.26 and higher.

The system entropy is collected from several non-deterministic hardware events and is mainly used by cryptographic applications. The virtual random number generator device (paravirtualized device) allows the host to pass through entropy to VM Guest operating systems. This results in a better entropy in the VM Guest.

To use Virtio RNG, add an `RNG` device in **virt-manager** or directly in the VM Guest's XML configuration:

```
<devices>
  <rng model='virtio'>
    <backend model='random'>/dev/random</backend>
  </rng>
</devices>
```

The host now should use `/dev/random`:

```
>lsof /dev/random
qemu-syst 4926 qemu      6r    CHR    1,8      0t0 8199 /dev/random
```

On the VM Guest, the source of entropy can be checked with:

```
>cat /sys/devices/virtual/misc/hw_random/rng_available
```

The current device used for entropy can be checked with:

```
>cat /sys/devices/virtual/misc/hw_random/rng_current  
virtio_rng.0
```

You should install the `rng-tools` package on the VM Guest, enable the service, and start it. Under SUSE Linux Enterprise Server 15, do the following:

```
#zypper in rng-tools  
#systemctl enable rng-tools  
#systemctl start rng-tools
```

6.4. Disable unused tools and devices

Per host, use one virtualization technology only. For example, do not use KVM and Xen on the same host. Otherwise, you may find yourself with a reduced amount of available resources, increased security risk and a longer software update queue. Even when the amount of resources allocated to each of the technologies is configured carefully, the host may suffer from reduced overall availability and degraded performance.

Minimize the amount of software and services available on hosts. Most default installations of operating systems are not optimized for VM usage. Install what you really need and remove all other components in the VM Guest.

Windows* Guest:

- Disable the screen saver
- Remove all graphical effects
- Disable indexing of hard disks if not necessary
- Check the list of started services and disable the ones you do not need
- Check and remove all unneeded devices
- Disable system update if not needed, or configure it to avoid any delay while rebooting or shutting down the host
- Check the Firewall rules
- Schedule backups and anti-virus updates appropriately
- Install the [VMDP](#) paravirtualized driver for best performance
- Check the operating system recommendations, such as on the [Microsoft Windows* 7 better performance](#) Web page.

Linux Guest:

- Remove or do not start the X Window System if not necessary
- Check the list of started services and disable the ones you do not need
- Check the OS recommendations for kernel parameters that enable better performance
- Only install software that you really need
- Optimize the scheduling of predictable tasks (system updates, hard disk checks, etc.)

6.5. Updating the guest machine type

QEMU machine types define details of the architecture that are particularly relevant for migration and session management. As changes or improvements to QEMU are made, new machine types are added. Old machine types are still supported for compatibility reasons, but to use improvements, we recommend to always migrate to the latest machine type when upgrading.

Changing the guest's machine type for a Linux guest is transparent. For Windows* guests, we recommend creating a snapshot or backup of the guest—in case Windows* has issues with the changes it detects, and subsequently the user reverts to the original machine type the guest was created with.



Changing the machine type

Refer to *the section called “Changing the machine type”* for documentation.

7. VM Guest-specific configurations and settings



Note

This section applies to QEMU / KVM hypervisor only.

7.1. ACPI testing

The ability to change a VM Guest's state heavily depends on the operating system. It is important to test this feature before any use of your VM Guests in production. For example, most Linux operating systems disable this capability by default, so this requires you to enable this operation (normally through Polkit).

ACPI must be enabled in the guest for a graceful shutdown to work. To check if ACPI is enabled, run:

```
>virsh dumpxml VMNAME | grep acpi
```

If nothing is printed, ACPI is not enabled for your machine. Use `virsh edit` to add the following XML under `<domain>`:

```
<features>
  <acpi/>
</features>
```

If ACPI was enabled during a Windows Server* guest installation, it is not sufficient to turn it on in the VM Guest configuration only. For more information, see <https://support.microsoft.com/en-us/kb/309283>.

Regardless of the VM Guest's configuration, a graceful shutdown is always possible from within the guest operating system.

7.2. Keyboard layout

Though it is possible to specify the keyboard layout from a `qemu-system-ARCH` command, we recommend configuring it in the `libvirt` XML file. To change the keyboard layout while connecting to a remote VM Guest using `vnc`, edit the VM Guest XML configuration file. For example, to add an `en-us` keymap, add in the `<devices>` section:

```
<graphics type='vnc' port='-1' autoport='yes' keymap='en-us' />
```

Check the `vncdisplay` configuration and connect to your VM Guest:

```
>virsh vncdisplay sles15 127.0.0.1:0
```

7.3. Spice default listen URL

If no network interface other than `lo` is assigned an IPv4 address on the host, the default address on which the spice server listens will not work. An error like the following one will occur:

```
>virsh start sles15
error: Failed to start domain sles15
error: internal error: process exited while connecting to monitor: ((null):26929): Spice-
2015-08-12T11:21:14.221634Z qemu-system-x86_64: failed to initialize spice server
```

To fix this, you can change the default `spice_listen` value in `/etc/libvirt/qemu.conf` using the local IPv6 address `:::1`. The spice server listening address can also be changed on a per VM Guest basis, use `virsh edit` to add the `listen` XML attribute to the `graphics type='spice'` element:

```
<graphics type='spice' listen=':::1' autoport='yes' />>
```

7.4. XML to QEMU command line

Sometimes it could be useful to get the QEMU command line to launch the VM Guest from the XML file.

```
>virsh domxml-to-native① qemu-argv② SLE15.xml③
```

- ① Convert the XML file in domain XML format to the native guest configuration
- ② For the QEMU/KVM hypervisor, the format argument needs to be qemu-argv
- ③ Domain XML file to use

```
>sudo virsh domxml-to-native qemu-argv /etc/libvirt/qemu/SLE15.xml
LC_ALL=C PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin \
_QEMU_AUDIO_DRV=none /usr/bin/qemu-system-x86_64 -name SLE15 -machine \
pc-i440fx-2.3,accel=kvm,usb=off -cpu SandyBridge -m 4048 -realtime \
mlock=off -smp 4,sockets=4,cores=1,threads=1 -uuid 8616d00f-5f05-4244-97cc-86aaed8aea \
-no-user-config -nodefaults -chardev socket,id=charmonitor,path=/var/lib/libvirt/qemu/ \
-mon chardev=charmonitor,id=monitor,mode=control -rtc base=utc,driftfix=slew \
-global kvm-pit.lost_tick_policy=discard -no-hpet \
-no-shutdown -global PIIX4_PM.disable_s3=1 -global PIIX4_PM.disable_s4=1 \
-boot strict=on -device ich9-usb-ehci1,id=usb,bus=pci.0,addr=0x4.0x7 \
-device ich9-usb-uhci1,masterbus=usb.0,firstport=0,bus=pci.0,multifunction=on,addr=0x4 \
-device ich9-usb-uhci2,masterbus=usb.0,firstport=2,bus=pci.0,addr=0x4.0x1 \
-device ich9-usb-uhci3,masterbus=usb.0,firstport=4,bus=pci.0,addr=0x4.0x2 \
-drive file=/var/lib/libvirt/images/SLE15.qcow2,if=none,id=drive-virtio-disk0,format=qcow2 \
-device virtio-blk-pci,scsi=off,bus=pci.0,addr=0x6,drive=drive-virtio-disk0,id=virtio-disk0 \
-drive if=none,id=drive-ide0-0-1,readonly=on,format=raw \
-device ide-cd,bus=ide.0,unit=1,drive=drive-ide0-0-1,id=ide0-0-1 -netdev tap,id=hostnet0 \
-device virtio-net-pci,netdev=hostnet0,id=net0,mac=52:54:00:28:04:a9,bus=pci.0,addr=0x5 \
-chardev pty,id=charserial0 -device isa-serial,chardev=charserial0,id=serial0 \
-vnc 127.0.0.1:0 -device cirrus-vga,id=video0,bus=pci.0,addr=0x2 \
-device virtio-balloon-pci,id=balloon0,bus=pci.0,addr=0x5 -msg timestamp=on
```

7.5. Change kernel parameters at boot time

7.5.1. SUSE Linux Enterprise 11

To change the value for SLE 11 products at boot time, you need to modify your `/boot/grub/menu.lst` file by adding the `OPTION=parameter`. Then reboot your system.

7.5.2. SUSE Linux Enterprise 12 and 15

To change the value for SLE 12 and 15 products at boot time, you need to modify your `/etc/default/grub` file. Find the variable starting with `GRUB_CMDLINE_LINUX_DEFAULT` and add at the end `OPTION=parameter` (or change it with the correct value if it is already available).

Now you need to regenerate your `grub2` configuration:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

Then reboot your system.

7.6. Add a device to an XML configuration

To create a new VM Guest based on an XML file, you can specify the QEMU command line using the special tag `qemu:commandline`. For example, to add a `virtio-balloon-pci`, add this block at the end of the XML configuration file (before the `</domain>` tag):

```
<qemu:commandline>
  <qemu:arg value='-device' />
  <qemu:arg value='virtio-balloon-pci,id=balloon0' />
</qemu:commandline>
```

7.7. Adding and removing CPUs

Certain virtualization environments allow adding or removing CPUs while the virtual machine is running.

For the safe removal of CPUs, deactivate them first by executing

```
#echo 0 > /sys/devices/system/cpu/cpuX/online
```

Replace `X` with the CPU number. To bring a CPU back online, execute

```
#echo 1 > /sys/devices/system/cpu/cpuX/online
```

7.8. SCSI persistent reservation on a multipathed device

SCSI persistent reservations allow restriction of access to block devices in a shared storage setup. This avoids improper multiple parallel accesses to the same block device from software components on local or remote hosts, which could lead to device damage and data corruption.

Find more information on managing storage multipath I/O in *Chapter 18, Managing multipath I/O for devices*. Find more information about SCSI persistent reservations in *the section called “SCSI persistent reservations and `mpathpersist`”*.

For the virtualization scenario, QEMU's SCSI passthrough devices `scsi-block` and `scsi-generic` support passing guest persistent reservation requests to a privileged external helper program `qemu-pr-helper`. This needs to start before QEMU and creates a listener socket that accepts incoming connections for communication with QEMU.



Live migration scenario with multipathed devices

We recommend using the `multipath` alias instead of `wwid`. It is useful in the VM Guest live migration scenario, because it makes sure that the storage paths are identical between the source and destination hosts.

Find more information about `multipath` in *the section called “Setting aliases for multipath maps”*.

Procedure 10. Adding a SCSI persistent reservation in a VM Guest against the related multipathed device in the VM Host Server:

1. In the VM Host Server, create a multipath environment. For more information, refer to *the section called “Configuring the system for multipathing”* and *the section called “Multipath configuration”*.
2. In the VM Host Server, configure the `<reservations/>` sub-element of the `<source/>` element of the `<disk/>` element for the passed-through `lun` in your `libvirt` domain configuration. Refer to [libvirt Domain XML format](#).
3. In the VM Guest, install the `sg3_utils` package and reserve the SCSI disks on demand by using the `sg_persist` command.

Example 1. Practical example

1. In the VM Host Server, verify that the `multipathd.service` is running, and that a multipathed disk exists and is named, for example, `storage1`.

```
>sudo systemctl status multipathd.service
● multipathd.service - Device-Mapper Multipath Device Controller
   Loaded: loaded (/usr/lib/systemd/system/multipathd.service; enabled; preset:
   Active: active (running) since Sat 2023-08-26 21:34:13 CST; 1 week 1 day ago
   TriggeredBy: ○ multipathd.socket
   Main PID: 79411 (multipathd)
   Status: "up"
   Tasks: 7
         CPU: 1min 43.514s
        CGroup: /system.slice/multipathd.service
                └─79411 /sbin/multipathd -d -s
>sudo multipath -ll
storage1 (36589cfc000000537c47ad3eb2b20216e) dm-6 TrueNAS,iSCSI Disk
size=50G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  `-- 16:0:0:0 sdg 8:96 active ready running
`+- policy='service-time 0' prio=50 status=enabled
   `-- 17:0:0:0 sdh 8:112 active ready running
```

2. In the VM Host Server, add a `<disk/>` element in the VM Guest configuration file by running `virsh edit`.


```
<disk type='block' device='lun'❶>
  <driver name='qemu' type='raw'/>
  <source dev='/dev/mapper/storage1'>
    <reservations❷ managed='yes'❸>
  </source>
  <target dev='sda' bus='scsi'/>
  <address type='drive' controller='0' bus='0' target='0' unit='0'❹>
</disk>
```

- ❶ To support persistent reservations, the disks must be marked as `lun` with type `block` so that QEMU does SCSI passthrough.
- ❷ If present, it enables persistent reservations for SCSI-based disks. The element has one mandatory attribute `managed` with accepted values `yes` and `no`.
- ❸ If `managed` is `yes`, `libvirt` prepares and manages any resources needed.

When the value of the attribute `managed` is `no`, then the hypervisor acts as a client and the path to the server socket must be provided in the child element `source`, which currently accepts only the following attributes:

type The only valid option is `unix`.

path The path to the server socket.

mode The role of the hypervisor. Valid is `client`.

- ❹ Verify that the virtio-scsi HBA that the disk attaches already exists and has available units (the maximum count of units per virtio-scsi HBA is 7). Otherwise, you need to manually add a virtio-scsi HBA to avoid automatically adding the LSI HBA by `libvirt`. For example:

```
<controller type='scsi' index='0' model='virtio-scsi'>
  <address type='pci' domain='0x0000' bus='0x03' slot='0x00' function='0x0'/>
</controller>
```

```
>sudo virsh domblklist sles15sp5
Target    Source
-----
vda       /mnt/images/sles15sp5/disk0.qcow2
sda       /dev/mapper/storage1
```

3. In the VM Host Server, start the VM Guest. `libvirt` launches a `qemu-pr-helper` instance as the server role for the VM Guest `sles15sp5`, then launches the VM Guest `sles15sp5` as the client role.

```
>sudo virsh start sles15sp5
Domain 'sles15sp5' started
>sudo virsh list
  Id    Name         State
  ----
   4    sles15sp5    running
>sudo ps -eo pid,args | grep -v grep | grep qemu-pr-helper
37063 /usr/bin/qemu-pr-helper -k /var/lib/libvirt/qemu/domain-4-sles15sp5/pr-hel
```

```
>sudo virsh dumpxml sles15sp5 | grep -A11 "<disk type='block' device='lun'">
<disk type='block' device='lun'>
  <driver name='qemu' type='raw'/>
  <source dev='/dev/mapper/storage1' index='1'>
    <reservations managed='yes'>
      <source type='unix' path='/var/lib/libvirt/qemu/domain-4-sles15sp5/pr-helper0'
    </reservations>
  </source>
  <backingStore/>
  <target dev='sda' bus='scsi'/>
  <alias name='scsi0-0-0-0'/>
  <address type='drive' controller='0' bus='0' target='0' unit='0'/>
</disk>
```

4. In the VM Guest, reserve the scsi disk, for example, sda, with the key 123abc.

```
>lsblk
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0   50G  0 disk
vda        253:0    0   20G  0 disk
├─vda1     253:1    0    8M  0 part
├─vda2     253:2    0    2G  0 part [SWAP]
└─vda3     253:3    0   18G  0 part /
>sudo sg_persist --verbose --out --register --param-sark=123abc /dev/sda
  inquiry cdb: [12 00 00 00 24 00]
  TrueNAS    iSCSI Disk          0123
  Peripheral device type: disk
  Persistent reservation out cdb: [5f 00 00 00 00 00 00 00 18 00]
PR out: command (Register) successful
>sudo sg_persist --verbose --in -k /dev/sda
  inquiry cdb: [12 00 00 00 24 00]
  TrueNAS    iSCSI Disk          0123
  Peripheral device type: disk
  Persistent reservation in cdb: [5e 00 00 00 00 00 00 20 00 00]
PR generation=0x5, 2 registered reservation keys follow:
  0x123abc
  0x123abc
```

5. In the VM Guest, release the sda disk with the key 123abc.

```
>sudo sg_persist --verbose --out --clear --param-rk=123abc /dev/sda
  inquiry cdb: [12 00 00 00 24 00]
  TrueNAS    iSCSI Disk          0123
  Peripheral device type: disk
  Persistent reservation out cdb: [5f 03 00 00 00 00 00 00 18 00]
PR out: command (Clear) successful
>sudo sg_persist --verbose --in -k /dev/sda
  inquiry cdb: [12 00 00 00 24 00]
  TrueNAS    iSCSI Disk          0123
  Peripheral device type: disk
  Persistent reservation in cdb: [5e 00 00 00 00 00 00 20 00 00]
PR generation=0x6, there are NO registered reservation keys
```

8. More information

- [Increasing memory density using KSM](#)
- [linux-kvm.org KSM](#)
- [KSM's kernel documentation](#)
- [ksm - dynamic page sharing driver for Linux v4](#)
- [Memory Ballooning](#)

- [libvirt virtio](#)
- [BFQ \(Budget Fair Queueing\)](#)
- [Documentation for sysctl](#)
- [LWN Random Number](#)
- [Kernel Parameters](#)
- [Huge pages Administration \(Mel Gorman\)](#)
- [kernel hugetlbpage](#)

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